THESIS ASSIGNMENT

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Class: Renewable energy and power system 01 – K67

Thesis title: *Resilience-Oriented Peer-to-Peer Energy Trading in Multi-Microgrids via Network-Constrained Distributed Predictive Control*

I. Review the state-of-the-art literature on microgrids and Peer-to-Peer (P2P) energy trading, identifying gaps in physical network constraints and resilient operations.

II. Develop the system architecture and mathematical formulation for the multi-microgrid energy management problem, strictly considering physical grid constraints.

III. Design a distributed distributed coordination framework to facilitate secure energy trading and sharing across interconnected microgrids without compromising individual operational privacy.

IV. Propose a real-time dynamic scheduling strategy to proactively manage energy resources and mitigate the impacts of extreme fault events.

V. Simulate and analyze the proposed framework on multi-microgrid testbeds, comprehensively evaluating economic efficiency, voltage stability, and overall system resilience.

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Supervisor

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